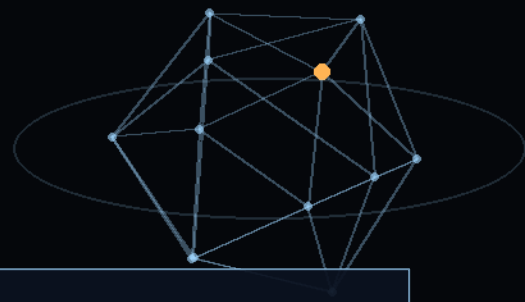
A Story: The Friday Question



THE QUESTION

Friday, 6 pm

The duty manager asks: are we meeting the departure immigration standard in Terminal 1?

THE CHATBOT

Answers from memory

It sounds sure, but it guesses the rule and invents a number. Nothing is checked.

THE ANALYST

Checks, then answers

Looks up the policy for the limit, asks the database for the real wait, then compares.

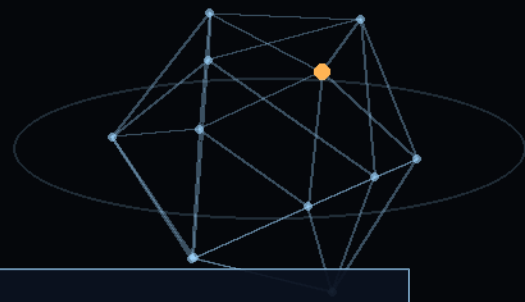
THE AGENT

Works like the analyst

Thinks, uses a tool, reads the result, and repeats until the evidence is enough.

An agent is a model that works like a careful analyst.

ReAct: Reason, Act, Observe



1 | THOUGHT

Reason

The model thinks in words. What do I know? What is missing? Which tool should I use next?

2 | ACTION

Act

It calls a tool with an input: a plain question for RAG, or a SQL query for the database.

3 | OBSERVATION

Observe

The tool result comes back. The model reads it and adds it to what it knows.

THE LOOP

Repeat until there is enough

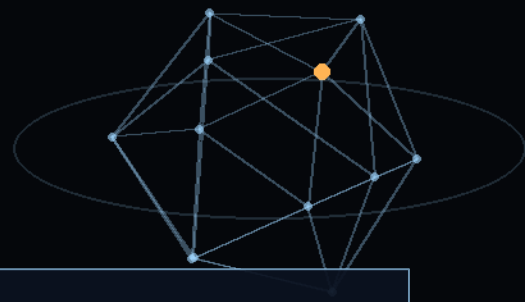
Thought, action, observation. Again and again. When the evidence is enough, the model writes the final answer.

WHY IT WORKS | YAO ET AL., 2022

Reasoning guides acting

The thought chooses the tool. The result corrects the next thought. Less guessing, more checking.

Worked Example: The Friday Question



THOUGHT 1

I need the rule first

Is Terminal 1 compliant? I must know the limit before I compare.

ACTION 1 | airport_policy

Ask the documents

Question: what is the maximum average wait for departure immigration?

OBSERVATION 1

The rule

15 minutes, and at least 9 counters in Terminal 1 at peak [GIA-POL-01, 2.3 and 3.2].

THOUGHT 2

Now I need the real numbers

Friday evening, 17:00 to 22:59, Terminal 1. Wait and counters open.

ACTION 2 | airport_database

Ask the database

A SELECT query: average wait and average counters open by weekday.

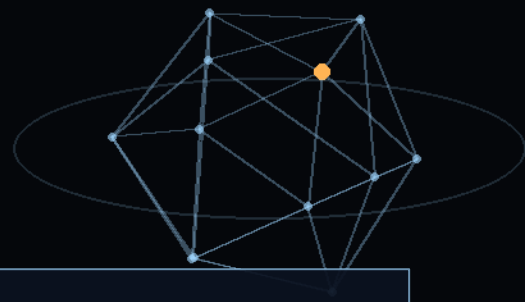
OBSERVATION 2

The data

Friday: about 36 minutes and 5 counters. Other days: about 9 to 10 minutes and 9 counters.

FINAL ANSWER: No. About 36 minutes against a 15-minute limit, with 5 counters instead of 9.

Different Types of Agents



REACT AGENT

Thought, action, observation

The model writes its reasoning as text, then names the tool. Works even without built-in tool calling. LlamaIndex: ReActAgent.

FUNCTION-CALLING AGENT

What we built

The model returns a structured tool call. Same loop, less text to parse. LlamaIndex: FunctionAgent with Gemini.

PLAN AND EXECUTE

Plan first, then run

The model writes the whole plan, then runs each step. Good for long, predictable tasks.

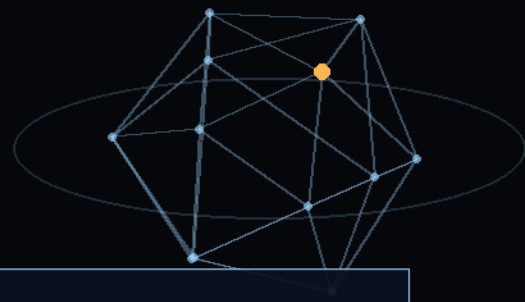
MULTI-AGENT

A team of specialists

One agent for policy, one for data, one to combine. More power, more to control.

Start with one agent and two tools. Add more only when you need it.

Why It Matters: The Value



GROUNDED

Fewer guesses

Answers come from your documents and your data, not from the model's memory.

FASTER

One question, two sources

No hunting through PDFs. No writing SQL by hand. You ask in plain language.

TRACEABLE

You can check it

Every answer shows the tools used and the sources, so a person can verify it.

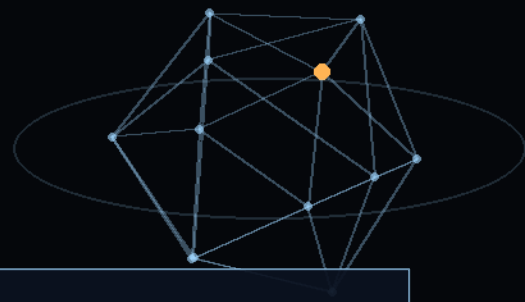
SAFE

Controlled by design

Read-only database, SELECT only, a row limit, and a person in the loop.

The value is in questions that need more than one source.

From RAG to an Agent: Two Tools, One Decision



THE QUESTION

A manager asks

Is Terminal 1 meeting the departure immigration standard on Friday evenings?

THE AGENT

Gemini 3.5 Flash

Reads the question and the tool descriptions. Chooses which tool to call, in which order, and when it has enough to answer.

TOOL 1 | RAG

airport_policy

Searches the 2 PDFs. The RULE: 15 minutes, 9 counters, escalation levels.

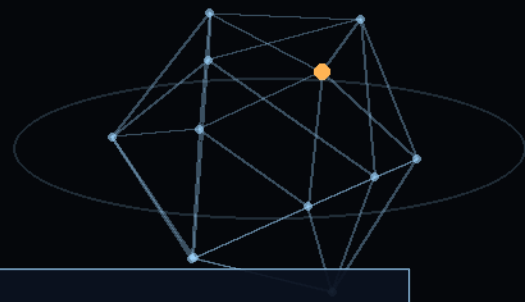
TOOL 2 | SQL

airport_database

Runs a read-only SELECT on airport_ops.db. The NUMBERS: 36 minutes, 5 counters.

One tool alone cannot answer. The rule is in the PDFs. The reality is in the database.

How the Agent Decides: The Loop



STEP 1

Think

Read the question. Which tool could help?

STEP 2

Call a tool

Pick the tool and write its input: a question, or a SQL query.

STEP 3

Read the result

The tool answer comes back as text. Is it enough?

STEP 4

Answer or repeat

Write the answer, or call another tool first.

THE KEY IDEA

The description is how it chooses

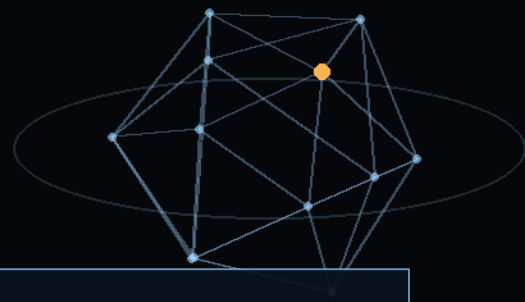
The agent never sees our code. It only reads each tool's name and description. A vague description gives wrong routing, so write it like a short job brief.

KEEP IT SAFE

Small, read-only tools

Database opened read-only. One SELECT only. At most 30 rows returned. A rule prompt says: I could not find this.

Lab: Build the Agent in Colab



STEPS 1 TO 2

Install and connect

Install the libraries. Key from Colab Secrets. Gemini 3.5 Flash, thinking level low.

STEP 3

Upload the files

The 2 airport PDFs and airport_ops.db, selected in one go.

STEP 4

Tool 1: RAG

Day 2 index: chunks 300, overlap 30, top_k 3, plus the rule prompt. Wrap as airport_policy.

STEP 5

Tool 2: SQL

A read-only SELECT function. The description lists tables and columns. Wrap as airport_database.

STEP 6

Create the agent

FunctionAgent with both tools. Ask 5 questions: rule, numbers, both, both, not found.

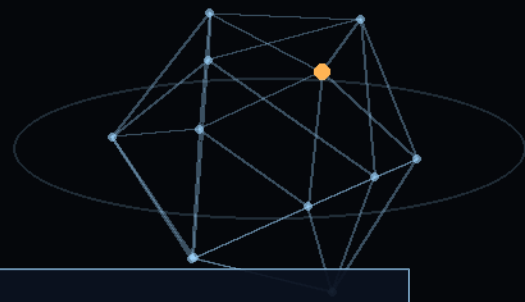
STEP 7

Gradio app

Answer box plus a box showing which tools the agent used.

Your turn: break the routing, try to delete data, then ask your own questions.

Putting It All Together



DAY 1 | PROMPTS AND DATA

Understand and ask

Explore the airport database in Python. Read the documents. Call Gemini in code, with a system prompt and context.

DAY 2 | RAG

Answer from documents

Embeddings, chunks 300 with overlap 30, top_k 3, and a rule prompt. A Gradio app that shows its sources.

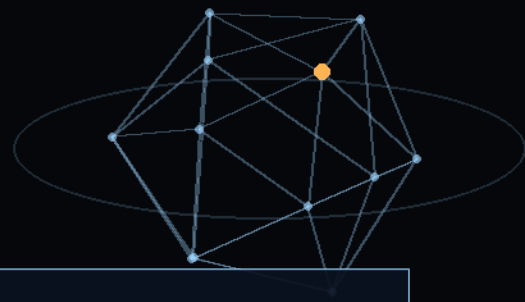
DAY 3 | AGENT

Choose the right tool

A policy tool (RAG) and a database tool (SQL). The agent picks, combines, and shows which tools it used.

Prompt + documents + data = one assistant that knows the rule and the number.

Next Steps



THIS WEEK

Finish the assessment

Play the post quiz and fill in the short end-of-course form.

THIS WEEK

Keep your key safe

Colab Secrets only. Never share it. If it is exposed, delete it and create a new one.

NEXT 30 DAYS

Pick one work use case

Choose a small, low-risk question that needs a document and a table.

NEXT 30 DAYS

Test before you trust

Write 5 questions with known answers. Check every answer against the source.

AFTER THAT

Add one tool at a time

A new table or a new document. Keep the database read-only.

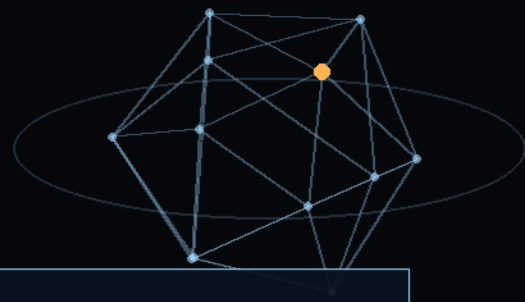
AFTER THAT

Share it safely

Check data sensitivity first. Keep a person in the loop before anyone relies on it.

Start small. Measure it. Keep a person in the loop.

Quick Evaluation



QUESTION 1

Relevant

The content is useful for my work.

QUESTION 2

Clear

The explanations and labs were easy to follow.

QUESTION 3

Practical

I can apply what I learned in my job.

QUESTION 4

Hands-on

The Colab notebooks worked well for me.

HOW TO ANSWER

Rate each from 1 to 5

1 = strongly disagree, 3 = neutral, 5 = strongly agree.

QUESTION 5 | OPTIONAL

One thing to improve

Write one sentence. What should we change next time?

Thank you. Your feedback shapes the next session.